(a) Single particle problem

$$H_1 = \begin{pmatrix} \varepsilon & -t \\ -t & \varepsilon \end{pmatrix} \Rightarrow \det\left((\varepsilon-\lambda)^2 - t^2\right) = 0$$

$$\lambda = \varepsilon \pm t$$

Eigenvalues

$$\begin{pmatrix} \varepsilon-\lambda & -t \\ -t & \varepsilon-\lambda \end{pmatrix}\begin{pmatrix} a \\ b \end{pmatrix} = 0$$

$$\lambda_1 = \varepsilon + t \Rightarrow (-t)\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}\begin{pmatrix} a \\ b \end{pmatrix} = 0 \Rightarrow a+b=0$$

$$\Rightarrow \psi_1 = \frac{1}{\sqrt{2}}\left(|R\rangle - |R'\rangle\right)$$

$$\lambda_2 = \varepsilon - t \Rightarrow t\begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}\begin{pmatrix} a \\ b \end{pmatrix} = 0 \Rightarrow a-b=0$$

$$\Rightarrow \psi_2 = \frac{1}{\sqrt{2}}\left(|R\rangle + |R'\rangle\right)$$

$\Rightarrow$ lowest energy state $\lambda = \varepsilon - t$ with symmetric spatial wave function $\psi = \dfrac{|R\rangle + |R'\rangle}{\sqrt{2}}$

(b) Independent electron approximation: every electron is on site $|R\rangle$ or $|R'\rangle$ with probability $1/2$

$\Rightarrow$ probability of <u>two of</u> them at $|R\rangle$ or $|R'\rangle$ is $\frac{1}{2}\cdot\frac{1}{2} + \frac{1}{2}\cdot\frac{1}{2} = \frac{1}{2}$.

$$\Rightarrow E = 2(\varepsilon - t) + \frac{1}{2}U$$

(c) $\phi_0 = \frac{1}{\sqrt{2}}\left(|R\rangle|R'\rangle + |R'\rangle|R\rangle\right)$, $\phi_1 = |R\rangle|R\rangle$

$\phi_2 = |R'\rangle|R'\rangle$

(c) Wave function of two independent electrons = product of their wave functions

$$\Rightarrow \quad \frac{1}{\sqrt{2}}\left(|R\rangle_1 + |R'\rangle_1\right)\frac{1}{\sqrt{2}}\left(|R\rangle_2 + |R'\rangle_2\right) =$$

$$= \frac{1}{2}\left(|R\rangle_1|R\rangle_2 + |R\rangle_1|R'\rangle_2 + |R'\rangle_1|R\rangle_2 + |R'\rangle_1|R'\rangle_2\right)$$

$$= \frac{1}{\sqrt{2}}\phi_0 + \frac{1}{2}(\phi_1 + \phi_2)$$

as in (32.36)

Now $H = h_1 + h_2 + V_{12}$.

For example $\langle\phi_0|H|\phi_0\rangle = \frac{1}{2}\left(\langle RR'| + \langle R'R|\right) H\left(|RR'\rangle + |R'R\rangle\right)$

$$= \frac{1}{2}\Big(\langle RR'|h_1+h_2|RR'\rangle + \langle R'R|h_1+h_2|R'R\rangle$$
$$+ \langle RR'|h_1+h_2|R'R\rangle + \langle R'R|h_1+h_2|RR'\rangle\Big)$$

$$= \frac{1}{2}\Big(\varepsilon + \varepsilon + \varepsilon + \varepsilon + \underbrace{\langle R|h_1|R'\rangle}_{-t}\underbrace{\langle R'|R\rangle}_{=0}\Big) = 2\varepsilon$$

$$\langle\phi_0|H|\phi_1\rangle = \frac{1}{\sqrt{2}}\langle RR' + R'R|h_1+h_2|RR\rangle =$$

$$= \frac{1}{\sqrt{2}}\Big(\langle R|h_1|R\rangle\cancel{\langle R'|R\rangle} + \underbrace{\langle R|R\rangle}_{1}\underbrace{\langle R'|h_2|R\rangle}_{-t}$$
$$+ \underbrace{\langle R'|h_1|R\rangle}_{-t}\langle R|R\rangle + \underbrace{\langle R'|R\rangle}_{=0}\langle R|h_2|R\rangle\Big)$$

$$= \frac{1}{\sqrt{2}}(-2t) = -\sqrt{2}\,t.$$

(d) Diagonalize full matrix

$$\det(\hat{H} - \lambda \hat{1}) = \det\begin{pmatrix} 2\varepsilon-\lambda & -\sqrt{2}t & -\sqrt{2}t \\ -\sqrt{2}t & 2\varepsilon+U-\lambda & 0 \\ -\sqrt{2}t & 0 & 2\varepsilon+U-\lambda \end{pmatrix}$$

$$= (2\varepsilon+U-\lambda)\left[(2\varepsilon-\lambda)(2\varepsilon+U-\lambda) - 2t^2\right] - 2t^2(2\varepsilon+U-\lambda)$$

$$= (2\varepsilon+U-\lambda)\left[(2\varepsilon-\lambda)(2\varepsilon+U-\lambda) - 4t^2\right] = 0$$

$$\lambda = 2\varepsilon+U, \quad (2\varepsilon-\lambda)^2 + U(2\varepsilon-\lambda) - 4t^2 = 0$$

$$x^2 + Ux - 4t^2 = 0, \quad x = 2\varepsilon - \lambda$$

$$x = -\frac{U}{2} \pm \sqrt{\frac{U^2}{4} + 4t^2}$$

$$\Rightarrow \lambda = 2\varepsilon - x = 2\varepsilon + \frac{U}{2} \pm \sqrt{\frac{U^2}{4} + 4t^2}$$

$\Rightarrow$ ground state $E_0 = 2\varepsilon + \frac{1}{2}U - \sqrt{4t^2 + \frac{1}{4}U^2}$

limits: $U \to 0 \Rightarrow E_0 \Rightarrow 2\varepsilon - 2t = 2(\varepsilon - t)$ Independent electrons

$$U \gg t \Rightarrow E_0 = 2\varepsilon + \frac{1}{2}U - \frac{U}{2}\sqrt{1 + \frac{16t^2}{U^2}} =$$

$$\hat{=} 2\varepsilon + \frac{1}{2}U - \frac{1}{2}U\left(1 + \frac{8t^2}{U^2}\right) = 2\varepsilon - \frac{4t^2}{U} < 2\varepsilon$$

Thus for strong interaction $U \gg t$ the ground state energy approaches HL approximation value ($H_{00} = 2\varepsilon$) from below.

(c) G.S. wave function

$$\begin{pmatrix} -\frac{1}{2}u+\sqrt{4t^2+\frac{1}{4}u^2} & -\sqrt{2}t & -\sqrt{2}t \\ -\sqrt{2}t & \frac{1}{2}u+\sqrt{4t^2+\frac{1}{4}u^2} & 0 \\ -\sqrt{2}t & 0 & \frac{1}{2}u+\sqrt{4t^2+\frac{1}{4}u^2} \end{pmatrix}\begin{pmatrix} a \\ b \\ c \end{pmatrix}=0$$

$\Rightarrow \quad b=c$ (from last two equations)

$$\left(-\frac{1}{2}u+\sqrt{4t^2+\frac{1}{4}u^2}\right)a-2\sqrt{2}t\,b=0$$

$$b=\frac{1}{2\sqrt{2}}\left(-\frac{u}{2t}+\sqrt{4+\frac{u^2}{4t^2}}\right)a$$

$$=\frac{1}{\sqrt{2}}\,a\left(\sqrt{1+\frac{u^2}{16t^2}}-\frac{u}{4t}\right)$$

$$\Rightarrow \quad \psi=a\,\phi_0+b\,(\phi_1+\phi_2)$$

Normalization $\quad |a|^2+2|b|^2=1$

$$|a|^2+2\,\frac{a^2}{2}=1 \quad\Rightarrow\quad a=\frac{1}{\sqrt{2}}$$

$$\Rightarrow \quad \psi=\frac{1}{\sqrt{2}}\,\phi_0+\frac{1}{2}\left\{\sqrt{1+\left(\frac{u}{4t}\right)^2}-\frac{u}{4t}\right\}(\phi_1+\phi_2).$$

$$P=2\cdot\frac{1}{4}\left[\sqrt{1+\left(\frac{u}{4t}\right)^2}-\frac{u}{4t}\right]^2$$

$u\to 0 \Rightarrow \quad P\to\frac{1}{2}$

$$\frac{u}{t}\gg 1 \Rightarrow \quad P\to\frac{1}{2}\left[\frac{u}{4t}\sqrt{1+\left(\frac{4t}{u}\right)^2}-\frac{u}{4t}\right]^2=\frac{2t^2}{u^2}\to 0$$